

Part II finite-scale model sheet

MMST nearest-neighbour kinematics, the Daubechies $D4$ scaling map, and a fixed Wick-quartic convention

Lamport-form definition and consistency proof

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Scope fixed. The resolution limit is first formulated on one fixed spatial torus $\mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z})$. The lattice kinetic energy is the MMST nearest-neighbour Hamiltonian, the observable embeddings are induced by the orthonormal Daubechies $D4$ mask, and the interaction convention is

$$\lambda \varepsilon_N \sum_{x \in \Lambda_N} :\Phi_N(x)^4:_{c_{N,L}}.$$

No Daubechies–Galerkin kinetic energy, infinite-line Hamiltonian, or continuum Hamiltonian is identified with this operator. Spatial-cutoff and thermodynamic limits are separate limits and require separately stated hypotheses.

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1 Geometric and canonical data

⟨1⟩ 1 Fix the torus and the dyadic lattices.

Fix $L > 0$, choose $s_0 \in \mathbb{N}_0$, put $r_0 = 2^{s_0}$ and

$$\varepsilon_0 := \frac{L}{r_0}, \quad \varepsilon_N := 2^{-N} \varepsilon_0, \quad r_N := 2^N r_0. \quad (1.1)$$

The position and momentum lattices are

$$\Lambda_N := \{\varepsilon_N n : -r_N \leq n \leq r_N - 1\} \subset \mathbb{T}_L, \quad (1)$$

$$\Gamma_N := \left\{ \frac{\pi j}{L} : -r_N \leq j \leq r_N - 1 \right\}. \quad (1.2)$$

All position indices are taken modulo $2r_N$. There are exactly $2r_N$ sites and $2r_N \varepsilon_N = 2L$.

⟨1⟩ 2 Fix the Fourier normalization.

For $f : \Lambda_N \rightarrow \mathbb{C}$, define

$$(F_N f)(k) := \sum_{x \in \Lambda_N} f(x) e^{-ikx}, \quad k \in \Gamma_N. \quad (1.3)$$

Then

$$f(x) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} (F_N f)(k) e^{ikx}, \quad \sum_x |f(x)|^2 = \frac{1}{2r_N} \sum_k |(F_N f)(k)|^2. \quad (1.4)$$

For the real canonical coordinates below, set

$$\hat{q}_N := \varepsilon_N^{1/2} F_N q, \quad \hat{p}_N := \varepsilon_N^{1/2} F_N p. \quad (1.5)$$

$\langle 1 \rangle 3$ Fix the symplectic and Weyl conventions.

Let $h_{N,L} = \ell^2(\Lambda_N)$. Write every one-particle vector uniquely as

$$\xi = \varepsilon_N q + ip, \quad q, p : \Lambda_N \rightarrow \mathbb{R}. \quad (1.6)$$

The symplectic form is

$$\sigma_{N,L}((q, p), (q', p')) = \varepsilon_N \sum_{x \in \Lambda_N} (q(x)p'(x) - p(x)q'(x)). \quad (1.7)$$

The Weyl algebra $\mathcal{W}_{N,L}$ is generated by unitaries $W_N(\xi)$ satisfying

$$W_N(\xi)W_N(\eta) = e^{-i\sigma_{N,L}(\xi,\eta)/2} W_N(\xi + \eta). \quad (1.8)$$

In the regular Schrödinger representation,

$$W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}, \quad (2)$$

$$\Phi_N(q) = \varepsilon_N \sum_x q(x) \Phi_N(x), \quad \Pi_N(p) = \varepsilon_N \sum_x p(x) \Pi_N(x), \quad (1.9)$$

and therefore

$$[\Phi_N(x), \Pi_N(y)] = i\varepsilon_N^{-1} \delta_{xy} I. \quad (1.10)$$

2 The D4 observable embeddings

$\langle 2 \rangle 1$ Fix the low-pass mask, including its normalization.

Put

$$h_0 = \frac{1 + \sqrt{3}}{4\sqrt{2}}, \quad h_1 = \frac{3 + \sqrt{3}}{4\sqrt{2}}, \quad (3)$$

$$h_2 = \frac{3 - \sqrt{3}}{4\sqrt{2}}, \quad h_3 = \frac{1 - \sqrt{3}}{4\sqrt{2}}, \quad (2.1)$$

and $h_n = 0$ otherwise. Define

$$m_0(t) := 2^{-1/2} \sum_{n=0}^3 h_n e^{-int}. \quad (2.2)$$

With $z = e^{-it}$, direct substitution gives

$$m_0(t) = \frac{1}{8} ((1 + \sqrt{3}) + (3 + \sqrt{3})z + (3 - \sqrt{3})z^2 + (1 - \sqrt{3})z^3) \quad (4)$$

$$= \left(\frac{1+z}{2} \right)^2 \underbrace{\frac{(1 + \sqrt{3}) + (1 - \sqrt{3})z}{2}}_{=: D(t)}. \quad (2.3)$$

In particular $m_0(0) = 1$. Moreover

$$|D(t)|^2 = 2 - \cos t, \quad |m_0(t)|^2 = \cos^4(t/2)(2 - \cos t). \quad (2.4)$$

Writing $c = \cos t$, expand

$$|m_0(t)|^2 + |m_0(t + \pi)|^2 = \frac{(1 + c)^2(2 - c) + (1 - c)^2(2 + c)}{4} = 1. \quad (2.5)$$

Indeed, the two cubic polynomials are $2 + 3c - c^3$ and $2 - 3c + c^3$, whose sum is 4. Thus $0 \leq |m_0(t)| \leq 1$, and (2.5) is the quadrature-mirror identity used below.

$\langle 2 \rangle 2$ Define the one-step refinement map.

For $q, p \in \ell^2(\Lambda_N; \mathbb{R})$, define

$$(R_N^{N+1}(q, p))(y) := \sqrt{2} \sum_{x \in \Lambda_N} \sum_{n=0}^3 h_n(q(x), p(x)) \delta_{x+n\varepsilon_{N+1}}(y). \quad (2.6)$$

For $M < N$, set

$$R_M^N := R_{N-1}^N \circ \dots \circ R_M^{M+1}. \quad (2.7)$$

The QMF identity (2.5), equivalently $\sum_n h_n h_{n+2j} = \delta_{j0}$, gives by direct expansion

$$\sigma_{N+1,L}(R_N^{N+1}\xi, R_N^{N+1}\eta) = \sigma_{N,L}(\xi, \eta). \quad (2.8)$$

Hence the rule

$$\alpha_M^N(W_M(\xi)) := W_N(R_M^N \xi) \quad (2.9)$$

defines an injective unital $*$ -homomorphism and

$$\alpha_N^K \circ \alpha_M^N = \alpha_M^K. \quad (2.10)$$

$\langle 2 \rangle 3$ Record the exact support enlargement.

If q, p are supported in $X \subset \Lambda_M$, every site occurring in $R_M^N(q, p)$ has the form

$$x + \sum_{j=1}^{N-M} n_j \varepsilon_{M+j}, \quad x \in X, \quad n_j \in \{0, 1, 2, 3\}. \quad (2.11)$$

Consequently, before periodization,

$$\text{supp } R_M^N(q, p) \subset X + [0, 3\varepsilon_M(1 - 2^{-(N-M)})] \subset X + [0, 3\varepsilon_M]. \quad (2.12)$$

This is the support enlargement to be used in every local estimate.

$\langle 2 \rangle 4$ Fix the continuum observable embedding.

Let ϕ be the compactly supported orthonormal $D4$ scaling function, normalized by $\int \phi = 1$ and supported in $[0, 3]$. Let

$$\phi_{M,x}(y) := \varepsilon_M^{-1/2} \phi((y - x)/\varepsilon_M), \quad (2.13)$$

periodized on $\mathbb{T}_L$. Define

$$R_M^\infty(q, p) := \varepsilon_M^{1/2} \sum_{x \in \Lambda_M} (q(x) + ip(x)) \phi_{M,x}. \quad (2.14)$$

The scaling equation gives

$$R_N^\infty R_M^N = R_M^\infty. \quad (2.15)$$

Thus

$$\beta_M(W_M(\xi)) := W_{\text{ct}}(R_M^\infty \xi) \quad (2.16)$$

is an injective $*$ -homomorphism and

$$\beta_N \circ \alpha_M^N = \beta_M. \quad (2.17)$$

3 Nearest-neighbour Hamiltonian and Wick convention

⟨3⟩ 1 Fix the free operator.

Let $m > 0$. On $\mathcal{H}_{N,L} = L^2(\mathbb{R}^{2r_N}, d\varphi)$ take the Schrödinger realization of (1.10) and define on Schwartz space

$$H_{N,L}^{(0)} := \frac{\varepsilon_N}{2} \sum_{x \in \Lambda_N} \left[\Pi_N(x)^2 + m^2 \Phi_N(x)^2 + \varepsilon_N^{-2} (\Phi_N(x + \varepsilon_N) - \Phi_N(x))^2 \right]. \quad (3.1)$$

Its one-particle dispersion is exactly

$$\gamma_N(k) = \sqrt{m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2)}, \quad k \in \Gamma_N. \quad (3.2)$$

The free vacuum state is

$$\omega_{N,L}^{(0)}(W_N(\varepsilon_N q + ip)) = \exp \left[-\frac{1}{4(2r_N)} \sum_{k \in \Gamma_N} \left(\gamma_N(k)^{-1} |\hat{q}_N(k)|^2 + \gamma_N(k) |\hat{p}_N(k)|^2 \right) \right]. \quad (3.3)$$

⟨3⟩ 2 Fix the Wick covariance with no hidden additive constant.

Translation invariance gives

$$c_{N,L} := \omega_{N,L}^{(0)}(\Phi_N(x)^2) \quad (5)$$

$$= \frac{1}{2r_N} \sum_{k \in \Gamma_N} \frac{1}{2\varepsilon_N \gamma_N(k)}. \quad (3.4)$$

The finite-scale Wick polynomial is, by definition,

$$:\Phi_N(x)^4:_{c_{N,L}} := \Phi_N(x)^4 - 6c_{N,L} \Phi_N(x)^2 + 3c_{N,L}^2 I. \quad (3.5)$$

For comparison, after the thermodynamic limit at fixed N ,

$$c_N = \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{d\theta}{\sqrt{(m\varepsilon_N)^2 + 4\sin^2(\theta/2)}} = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1). \quad (3.6)$$

Equation (3.4), not (3.6), is the convention in the fixed-torus Hamiltonian.

⟨3⟩ 3 Fix the interacting Hamiltonian.

For $\lambda \geq 0$ and a function $g_N : \Lambda_N \rightarrow [0, 1]$, define

$$V_{N,L}(g_N) := \lambda \varepsilon_N \sum_{x \in \Lambda_N} g_N(x) : \Phi_N(x)^4 :_{c_{N,L}}, \quad (6)$$

$$K_{N,L}(g_N) := H_{N,L}^{(0)} + V_{N,L}(g_N). \quad (3.7)$$

The resolution problem treated in the companion documents uses $g_N \equiv 1$. The factor λ , rather than $\lambda/4!$, is part of the definition. The second line of (3.7) denotes the self-adjoint operator obtained from the closed quadratic form constructed below; it is not an assertion about an unspecified operator domain.

Proposition 3.1 (finite-volume operator and ground state). *For every $N, L, m > 0$, $\lambda \geq 0$, and $0 \leq g_N \leq 1$, the polynomial Schrödinger expression (3.7) has a lower-bounded closed quadratic-form realization $K_{N,L}(g_N)$. This realization has compact resolvent, and its lowest eigenvalue is simple.*

Proof in Lamport form

⟨1⟩ 1 Bound the negative Wick term.

For every $u \in \mathbb{R}$ and $a \geq 0$, completing the square gives

$$u^4 - 6au^2 + 3a^2 = (u^2 - 3a)^2 - 6a^2 \geq -6a^2. \quad (3.8)$$

Thus $V_{N,L}(g_N)$ is bounded below by a finite constant.

⟨1⟩ 2 Establish coercivity of the total potential.

Write $d = 2r_N$, $c = c_{N,L}$, and let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be the complete multiplication potential in (3.7). Its gradient contribution is nonnegative, and its remaining, coordinatewise part is

$$\varepsilon_N \sum_x f_x(\varphi_x) + 3\lambda \varepsilon_N c^2 \sum_x g_N(x), \quad f_x(u) := \lambda g_N(x) u^4 + \left(\frac{m^2}{2} - 6\lambda c g_N(x) \right) u^2. \quad (3.9)$$

If $\lambda = 0$, then $f_x(u) = m^2 u^2 / 2 \rightarrow \infty$ at every site. Suppose $\lambda > 0$. For each site with $g_N(x) > 0$, the polynomial f_x has positive leading coefficient and hence tends to $+\infty$ as $|u| \rightarrow \infty$; for each site with $g_N(x) = 0$, it is again $m^2 u^2 / 2$. Because the set of sites is finite, for every $A > 0$ there is $R_A < \infty$ such that

$$\max_x |\varphi_x| \geq R_A \implies \sum_x f_x(\varphi_x) \geq A. \quad (3.10)$$

Indeed, choose R_A so that $f_x(u) \geq A - \sum_{y \neq x} \inf_{v \in \mathbb{R}} f_y(v)$ whenever $|u| \geq R_A$, simultaneously for all x . Since $\|\varphi\| \rightarrow \infty$ implies $\max_x |\varphi_x| \rightarrow \infty$, (3.10) proves

$$U(\varphi) \longrightarrow +\infty \quad \text{as} \quad \|\varphi\| \longrightarrow \infty. \quad (3.11)$$

In the special case $g_N \equiv 1$, one also has the explicit inequality

$$\sum_x \varphi_x^4 \geq \frac{1}{2r_N} \left(\sum_x \varphi_x^2 \right)^2. \quad (3.12)$$

⟨1⟩ 3 Construct the self-adjoint operator by a closed form.

Choose $C > 0$ so that $W := U + C \geq 1$, which is possible by (3.8). On $L^2(\mathbb{R}^d)$, define

$$\mathcal{D}(\mathfrak{q}_C) := H^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d, W d\varphi), \quad (7)$$

$$\mathfrak{q}_C[\psi] := \frac{1}{2\varepsilon_N} \int_{\mathbb{R}^d} |\nabla \psi|^2 d\varphi + \int_{\mathbb{R}^d} W |\psi|^2 d\varphi. \quad (3.13)$$

This domain is dense because it contains $C_c^\infty(\mathbb{R}^d)$. It is complete for the norm $\|\psi\|_{\mathfrak{q}_C}^2 := \mathfrak{q}_C[\psi]$: if (ψ_j) is Cauchy, it converges in H^1 to some ψ , while $W^{1/2}\psi_j$ converges in L^2 to some v . A subsequence of ψ_j converges pointwise almost everywhere to ψ , and the same subsequence has a further subsequence for which $W^{1/2}\psi_j \rightarrow v$ pointwise almost everywhere. Hence $v = W^{1/2}\psi$, proving $\psi \in \mathcal{D}(\mathfrak{q}_C)$ and form-norm convergence. Thus $\mathfrak{q}_C$ is closed. The representation theorem for closed positive forms yields a unique positive self-adjoint operator $K_{N,L}(g_N) + C$; this defines $K_{N,L}(g_N)$. If $\psi, \zeta \in C_c^\infty(\mathbb{R}^d)$, integration by parts gives

$$\mathfrak{q}_C(\zeta, \psi) = \left\langle \zeta, \left(-\frac{1}{2\varepsilon_N} \Delta + U + C \right) \psi \right\rangle. \quad (3.14)$$

Consequently the form operator extends exactly the polynomial differential expression (3.7) on $C_c^\infty(\mathbb{R}^d)$.

⟨1⟩ 4 Prove compactness of the resolvent.

Let (ψ_j) be bounded in the form norm. Given $B > 0$, coercivity (3.11) supplies R_B such that $W(\varphi) \geq B$ for $\|\varphi\| \geq R_B$. Therefore

$$\int_{\|\varphi\| \geq R_B} |\psi_j|^2 d\varphi \leq B^{-1} \int W |\psi_j|^2 d\varphi \leq B^{-1} \sup_j \mathfrak{q}_C[\psi_j]. \quad (3.15)$$

On each ball, (ψ_j) is bounded in H^1 , so the Rellich compactness theorem gives an L^2 -convergent subsequence there. A diagonal subsequence, together with the uniform tail bound (3.15) and then $B \rightarrow \infty$, is Cauchy in $L^2(\mathbb{R}^d)$. Hence the form-domain inclusion into L^2 is compact. The resolvent $(K_{N,L}(g_N) + C)^{-1}$ maps L^2 boundedly into the form domain by the spectral theorem; composing it with the compact inclusion proves that this resolvent is compact.

⟨1⟩ 5 Prove simplicity of the lowest eigenvalue.

Compact resolvent and lower boundedness give a lowest eigenvalue. For $t > 0$, the Feynman–Kac formula gives the integral kernel

$$e^{-tK_{N,L}(g_N)}(\mathbf{x}, \mathbf{y}) = \left(\frac{\varepsilon_N}{2\pi t}\right)^{d/2} e^{-\varepsilon_N \|\mathbf{x} - \mathbf{y}\|^2 / (2t)} \mathbb{E}_{\mathbf{x}, \mathbf{y}}^t \left[\exp\left(-\int_0^t U(B_s) ds\right) \right]. \quad (3.16)$$

Here $\mathbb{E}_{\mathbf{x}, \mathbf{y}}^t$ is expectation for the Brownian bridge with diffusion coefficient ε_N^{-1} . Formula (3.16) follows first for bounded truncations of the continuous potential from the Trotter product formula and then for U by monotone form convergence. A bridge path is continuous almost surely, so its image is compact and the integral of the polynomial U along it is finite. Every factor on the right of (3.16) is therefore strictly positive. Thus $e^{-tK_{N,L}(g_N)}$ is positivity improving.

Choose a ground eigenfunction f . Since $|\nabla|f|| \leq |\nabla f|$ almost everywhere, the form satisfies $\mathfrak{q}_C[|f|] \leq \mathfrak{q}_C[f]$. The variational definition of the lowest eigenvalue forces equality, so $|f|$ is also a ground eigenfunction. Applying the positivity-improving semigroup to $|f| \neq 0$ shows that the normalized ground eigenfunction $\psi_0 := |f|/\|f\|$ is strictly positive almost everywhere.

Suppose that the ground eigenspace had dimension at least two. Since the operator commutes with complex conjugation, that eigenspace contains a real ground eigenfunction $h \neq 0$ orthogonal to ψ_0 . The strict positivity of ψ_0 and $\langle \psi_0, h \rangle = 0$ imply that both h_+ and h_- are nonzero. The strictly positive kernel in (3.16) then gives, almost everywhere,

$$|e^{-tK} h| < e^{-tK} |h|. \quad (3.17)$$

Pairing (3.17) with ψ_0 gives

$$e^{-tE_0} \langle \psi_0, |h| \rangle < \langle \psi_0, e^{-tK} |h| \rangle = e^{-tE_0} \langle \psi_0, |h| \rangle,$$

a contradiction. Hence the lowest eigenvalue is simple. □

Denote the normalized ground vector by $\Omega_{N,L,\lambda,g}$, subtract the ground energy, and set

$$\omega_{N,L,\lambda,g}(A) := \langle \Omega_{N,L,\lambda,g}, A \Omega_{N,L,\lambda,g} \rangle. \quad (3.18)$$

4 Scope ledger

Statement	Status	Exact reason
MMST nearest-neighbour free kinematics	PASS	Equations (1.1)–(1.10) and (3.1)–(3.3) fix the one-dimensional specialization.

Statement	Status	Exact reason
$D4$ wavelet observable embeddings	PASS	The mask, QMF identity, maps, and support enlargement are explicit in (2.1)–(2.17).
Finite-torus Wick quartic convention	PASS	Equations (3.4)–(3.7) fix the covariance, Wick polynomial, lattice measure, and coupling normalization.
Interacting resolution limit	FAIL	Defining the Hamiltonians supplies states but does not prove convergence of their pullbacks.
Infinite-volume/spatial-cutoff limit	FAIL	It is a distinct limit and is not a consequence of a fixed-torus definition.
Daubechies–Galerkin kinetic energy	FAIL	It is not the nearest-neighbour operator (3.1) and is outside this track.

References

- [1] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Commun. Math. Phys. **387** (2021), 1299–1367.